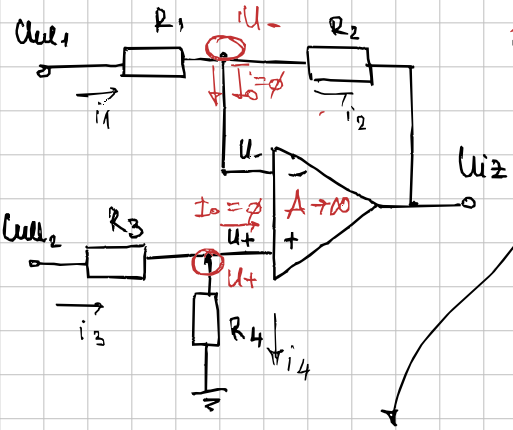
KROIS

1. Diferencijsko pojačalo (2. godina)

Idealno OP



kod idealnih operacijskih pojačala vrijedi

$i_1 = i_2 = \emptyset$

$R_{ul} = \infty$, $A_{vop} = \infty$

$A_{vop}(U_+ - U_-) = U_{iz}$

$$U_+ - U_- = \frac{U_{iz}}{A_{vop}} = \frac{U_{iz}}{\infty} = \emptyset$$

$U_+ = U_- \Rightarrow U$

$i_1 = i_2$, $i_3 = i_4$

$$U_{iz} = U_-\left(1 + \frac{R_2}{R_1}\right) - \frac{R_2}{R_1} \cdot U_{ul1}$$

$$U_-\left(\frac{1}{R_1} + \frac{1}{R_2}\right) = \frac{U_{iz}}{R_2} + \frac{U_{ul1}}{R_1}$$

$$U_+\left(\frac{1}{R_3} + \frac{1}{R_4}\right) = \frac{U_{ul2}}{R_3} \rightarrow U = \frac{U_{ul2}}{R_3}\left(\frac{1}{R_3} + \frac{1}{R_4}\right)^{-1} = \frac{U_{ul2}}{R_3}(R_3 \| R_4)$$

$$\frac{U_{ul2}}{\cancel{R_3}} \cdot \frac{\cancel{R_3} \cdot R_4}{R_3 + R_4} \cdot \frac{R_1 + R_2}{R_1 \cdot R_2} = \frac{U_{iz}}{R_2} + \frac{U_{ul1}}{R_1} \quad \Big/ \cdot R_2$$

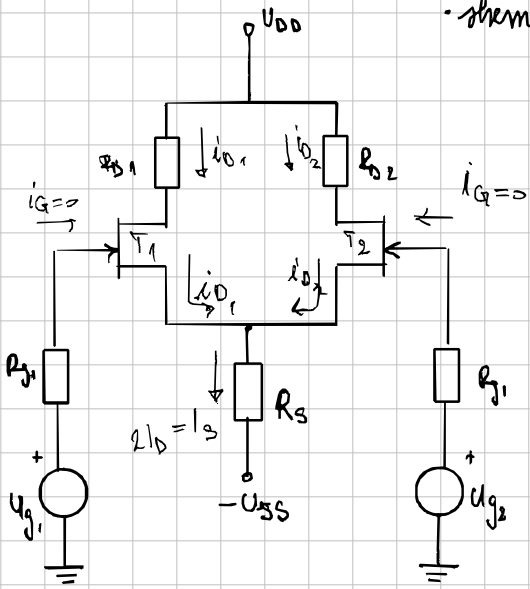
U_{iz} kao funkcija od U_{ul1}, U_{ul2}

$$U_{ul2} \cdot \frac{R_4}{R_3 + R_4} \cdot \frac{R_1 + R_2}{R_1} = U_{iz} + \frac{R_2}{R_1} U_{ul1} \rightarrow \boxed{U_{iz} = U_{ul2}\left(\frac{R_4}{R_3 + R_4}\right) \cdot \left(1 + \frac{R_2}{R_1}\right) - \frac{R_2}{R_1} U_{ul1}}$$

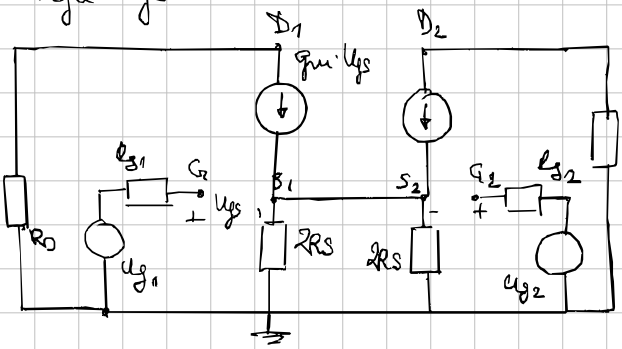
Grupa 1

1 Diferencijsko pojač. s uvodskim otpornikom

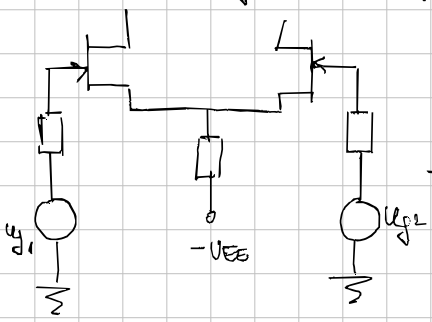
· sheme i izrazi za pojačanja
↳ koje je veće i zašto



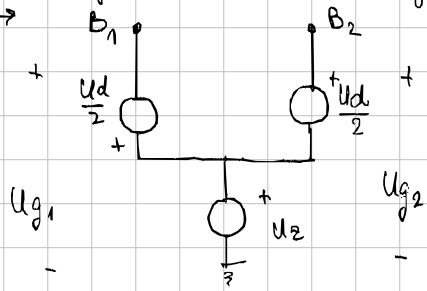
Pojačanje:



Diferencijska pojačala izvore U_{g1} i U_{g2} raščlanjuju na zajedničku U_Z komponentu koja predstavlja signal smetnje i diferencijsku komponentu koja je onaj „ajelohoni" signal



Umjesto U_{g1} i U_{g2} to gledamo:



$$U_{g1} = U_Z - \frac{U_d}{2}$$

$$U_{g2} = U_Z + U_d/2$$

što smo mogli dobiti iz $U_{g2} - U_{g1} = U_d$

$$\frac{U_{g2} + U_{g1}}{2} = U_Z$$

$$\frac{U_{g2}}{2} + \frac{U_{g1}}{2} = U_Z \quad , \quad U_{g2} - U_{g1} = U_d \;/ : 2$$

$$\frac{U_{g2}}{2} - \frac{U_{g1}}{2} = \frac{U_d}{2}$$

$$\Leftrightarrow \quad \underbrace{U_{g2} = U_Z + \frac{U_d}{2}}_{(+)} \; ; \quad \underbrace{U_{g1} = U_Z - \frac{U_d}{2}}_{(-)}$$

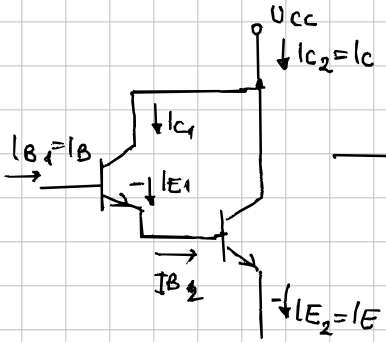
prema ovome računamo dinam. PRINCIP SUPER.

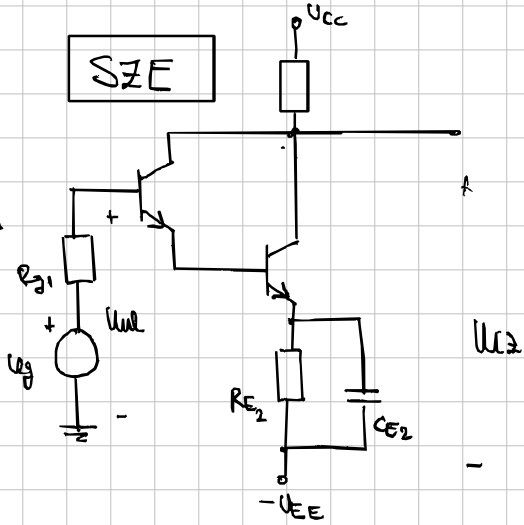
Pojačanja: $A_{vZ} < 1 \Rightarrow$ PRIGUŠUJEMO SMETNJE (istofazni signal)

$A_{vd} >> A_{vZ} \Rightarrow$ POJAČAVAMO DIFERENCIJSKI (onaj koji nas zanima)

2. Darlingtov spoj:

SAMOSTALAN:





$I_{B_1} = I_B \qquad I_{C_2} = I_C = I_{B_2} \cdot \beta_2$

$I_{E_2} = (1+\beta_2) \cdot I_{B_2} = I_E$

$I_{B_2} = I_{E_1} = (1+\beta_1)\, I_B$

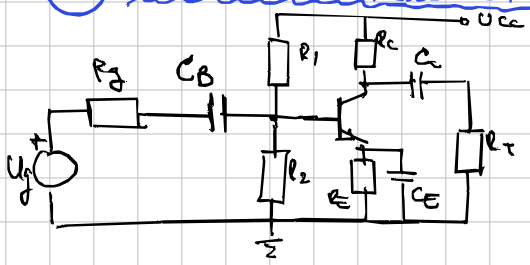
$I_C = I_B (1+\beta_1) \cdot \beta_2 \; ; \; \beta_1 ; \beta_2 >> 1 \longrightarrow \approx I_B \cdot \underbrace{\beta_1 \cdot \beta_2}_{\beta_{novi}} = I_C$

β je najvažniji parametar tranzistora → želimo što veći ali ovi su rijetki, skupi i teški za napraviti

→ spajanjem u Darlingtonov spoj povećavamo β spoja na iznos umnoška β_1 i β_2 korištenih tranzistora

→ izvana vidimo „normalan tranzistor" → spoje se tako da je jednako jednom T ali ovaj sada ima veći β pa je bolji

3. NF analiza neke sheme: +

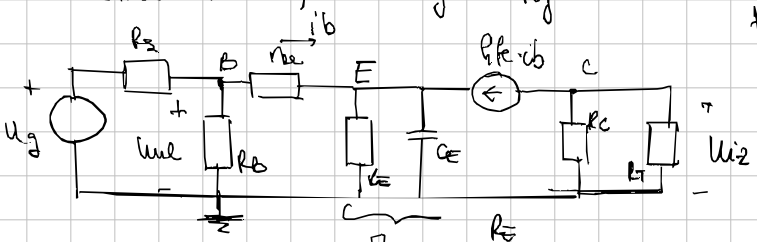


C_B i C_C su vezni kondenzatori

↳ spojeni su serijski sa signalom

C_E → mosni kondenzator → K.S. R_E

$\boxed{C_E}$ → sve ostale K.S (pretpostavimo da je frekv faktor da su C_B i C_C ušli u K.S., a C_E još nije)



$$Z_E = \frac{R_E}{j\omega C_E \cdot R_E + 1}$$

$\omega = 0$ → P.H.

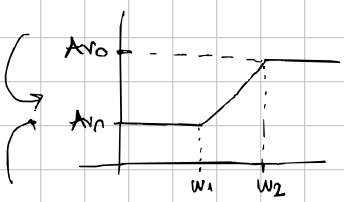
$Z = 0$ → K.S

$$Z_{CE} = \frac{1}{j\omega C_E}$$

$\omega \to 0$ $\quad Z_{CE} = \infty$ ---> odspojen C_E, još djeluje R_E → A_{vn}

$\omega \to \infty$ $\quad Z_{CE} = 0$ ---> C_E je K.S. → A_{vo}

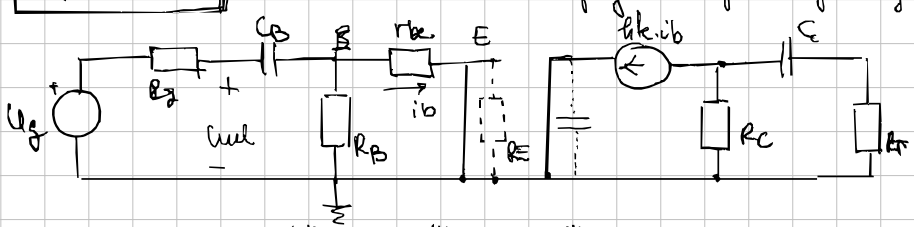
$$A_v(j\omega) = A_{vn} \cdot \frac{1 + \frac{j\omega}{\omega_1}}{1 + j\frac{\omega}{\omega_2}} \cdot \underbrace{\ldots}_{\text{ne znam još}}$$



$$\omega_1 = \frac{1}{T_1} = \frac{1}{C_E \cdot R_E}$$ → vlastita frekvencija

$$\omega_2 = \frac{1}{T_2} = \frac{1}{C_E \cdot (r_{be} + (1+h_{fe}) \cdot R_E)}$$

$\boxed{C_B \text{ i } C_C}$ sad C_E kratko spojamo po istoj logici

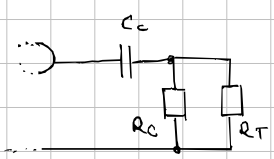


$$A_v(j\omega) = A_{vn} \cdot \frac{1 + j\frac{\omega}{\omega_1}}{1 + j\frac{\omega}{\omega_2}} \cdot \frac{j\frac{\omega}{\omega_3}}{1 + j\frac{\omega}{\omega_3}} \cdot \frac{j\frac{\omega}{\omega_4}}{1 + j\frac{\omega}{\omega_4}}$$

$$\omega_3 = \frac{1}{T_3} = (\text{K.S. } U_{ul} = 0)$$

→ $\omega_3 = \frac{1}{C_B \cdot R_B}$

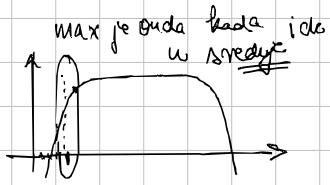
$$\omega_4 = \frac{1}{T_4} =$$



→ $\omega_4 = \frac{1}{C_C(R_C + R_T)}$

Svaki od veznih (serijski spojenih kond) ruši pojačanje na NF

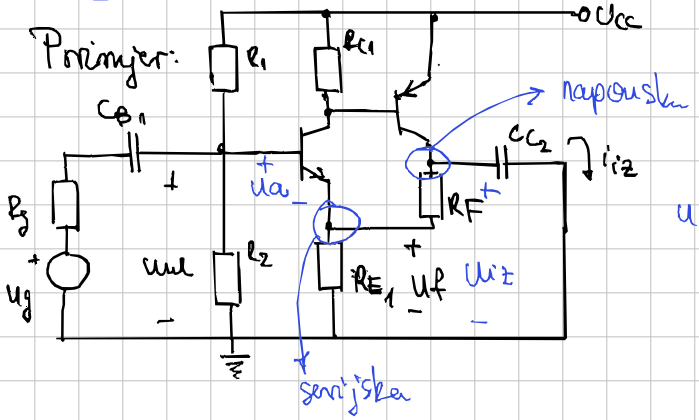
↳ C_G i C_B djeluju na $\omega \to 0$

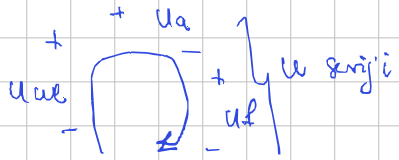


Donja granična frekv je $\max\{\omega_1, \omega_2, \omega_3, \omega_4\}$

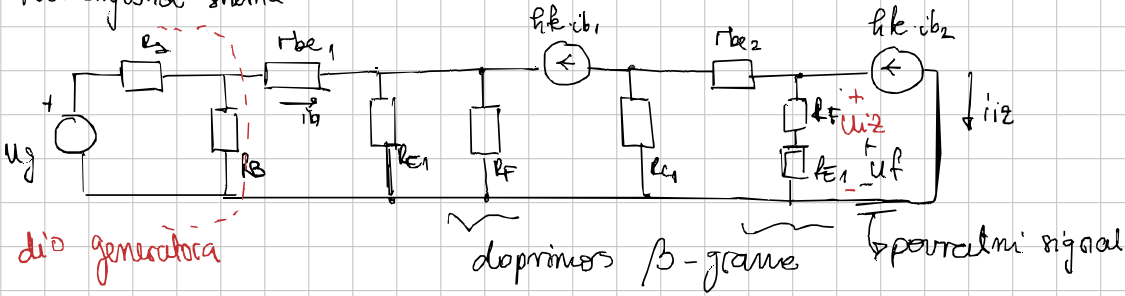
4. Povratna veza neka shema + (Naponsko-serijska) +

Primjer:





Nadomjesna shema



* U pojačalima se koristi negativna P.V.; iako smanjuje izlazni signal, ona i time pojačanje pojačala, P.V. osigurava

↳ stabilizira pojačanje; neosjetljivo na promjene parametra sklopa

↳ smanjuje nelinearna izobličenja

↳ povećava širinu frekvencijskog pojasa

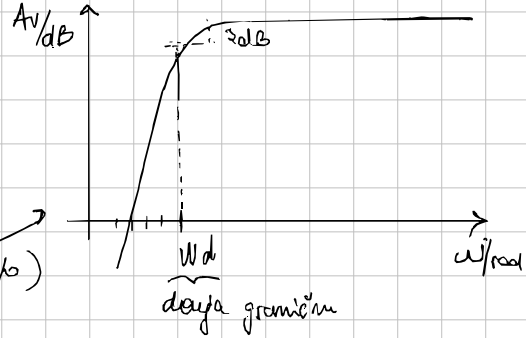
↳ Rul i Riz bliži idealnim pojačalima

$\beta = \frac{X_f}{X_{iz}}$ = koeficijent P.V.

⟹ uvođenjem P.V. pojačanje se po iznosu smanjuje u odnosu na bez, ALI POSTAJE STABILNIJE ⇝ ne ovisi o parametrima tranzistora

(4.1) Što utječe na donje graničnu frekvenciju?

Pokazati to matematički:



→ donje granične f su one na NF →

→ donja granična je najveća frekvencija (očito)

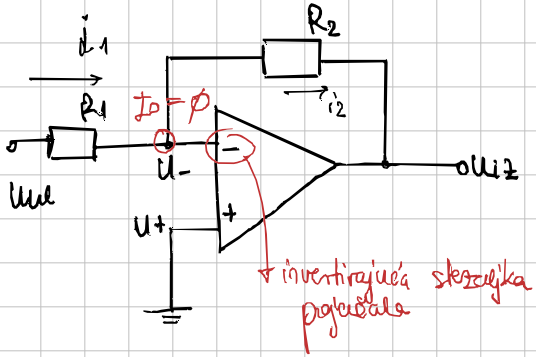
↳ ona na kojoj je Av palo za 3dB

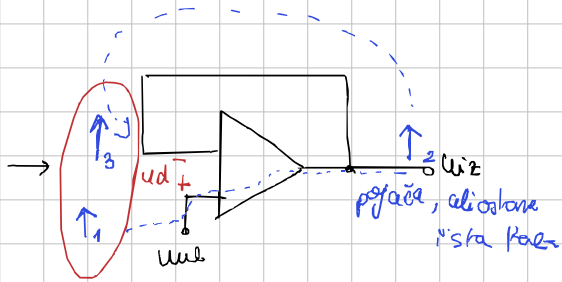
! Sljedeća mora biti dovoljno razmaknuta da ne utječu jedna na drugu

$\omega_d > 4\omega_{sljedeća}$! ; $\omega_d >> \ldots$ ostale → ω_d

→ ona ω pri kojoj je pojačanje $|A_v| = \frac{|A_{vo}|}{\sqrt{2}}$

5. Invertirajuće pojačalo



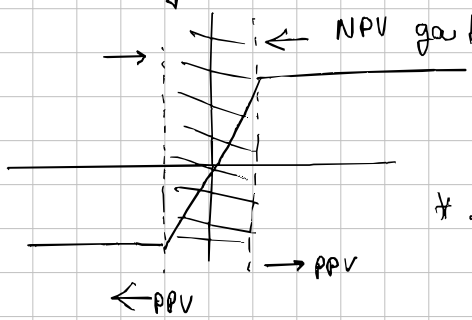


$U_{iz} = A_{vop}(\underbrace{U_+ - U_-}_{U_d})$

$i_3 > i_1 \longrightarrow U_d < 0$

negativna P.V.

→ povratna veza linearizira rad pojačala



← NPV ga tjera da ostane u tom području

↳ LINEARNO OP.

* za razliku od POZ (+) koja ga tjera u zasićenje

$$U_-\left(\frac{1}{R_1}+\frac{1}{R_2}\right) = \frac{U_{ul}}{R_1} + \frac{U_{iz}}{R_2}$$

A_{vop} (idealno) $\to \infty \to$ $U_+ = U_-$

$i_+ = i_- = \emptyset$

U_+ je uzemljeno

$U_- = U_+ = \emptyset$

$$\frac{U_{ul}}{R_1} = \frac{-U_{iz}}{R_2}$$

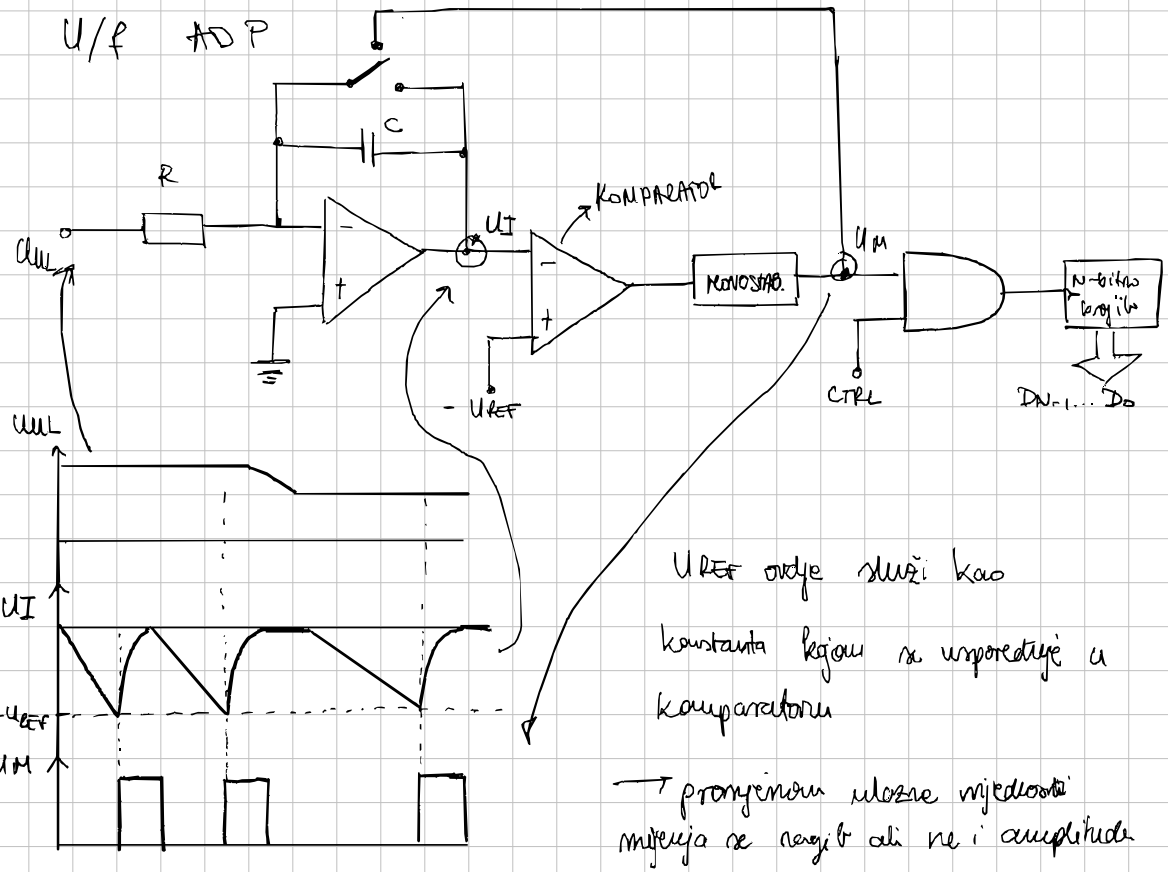
$$A_v = \frac{U_{iz}}{U_{ul}} \longrightarrow A_v = \frac{-R_2}{R_1}$$ idealno pojačanje

↳ to je posljedica $A_{vop} \to \infty$!

2. GRUPA

1. Nelinearnost, čemu služi UREF (ADP/DAP)

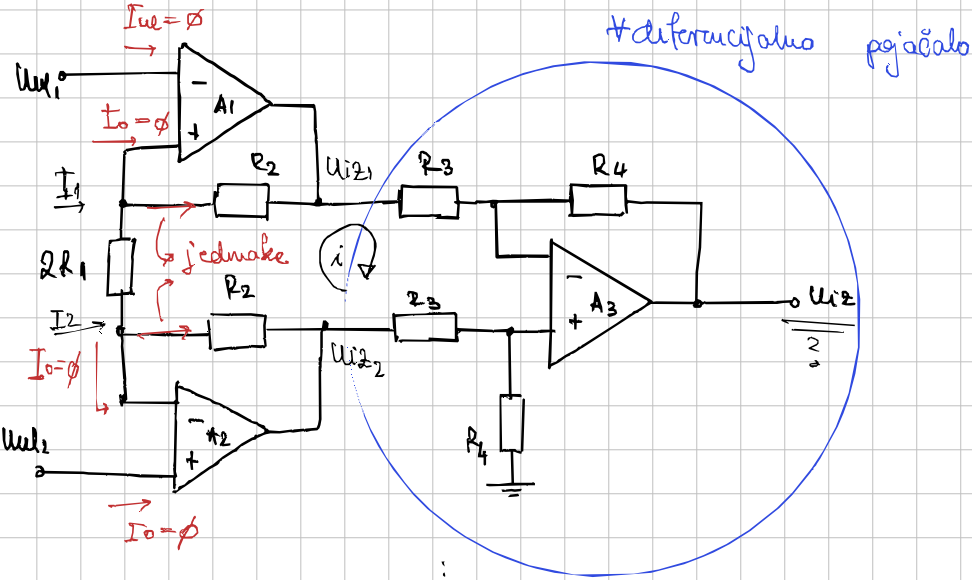
U/f ADP



UREF ovdje služi kao konstanta kojom se uspoređuje u komparatoru

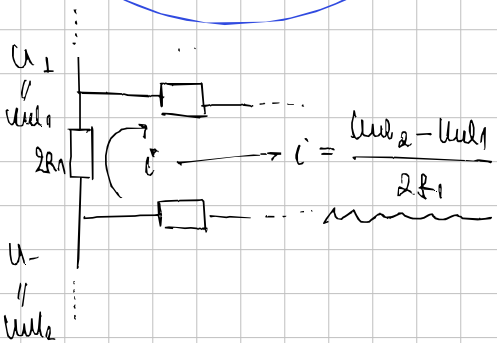
→ promjenom ulazne vrijednosti mijenja se razvoj ali ne i amplituda jer je -UREF konstantna veličina

2. Instrumentacijsko pojačalo – izvedi u_{iz}



diferencijalno pojačalo

$$i = \frac{u_{iz2} - u_{iz1}}{2R_2 + 2R_1}$$



$$i = \frac{u_{ul_2} - u_{ul_1}}{2R_1}$$

$$u_{iz} = \frac{R_4}{R_3}(u_{iz2} - u_{iz1}) \Rightarrow u_{iz} = \frac{R_4}{R_3} \cdot (u_{ul_2} - u_{ul_1}) \cdot \frac{2R_2 + 2R_1}{2R_1}$$

$$\boxed{u_{iz} = \frac{R_4}{R_3} \cdot \left(1 + \frac{R_2}{R_1}\right)(u_{ul_2} - u_{ul_1})}$$

3. i 4. isto

5. Što su kaskadna pojačala, čemu služe i koja je njihova podjela

→ samostalni spojevi nude ograničene zahtjeve

↳ npr Bip T SZE → A_V, A_I ≈100

R_{ul} << , R_{iz} >> , veća pojačanja ne mogu se postići drugim samostalnim spojevima

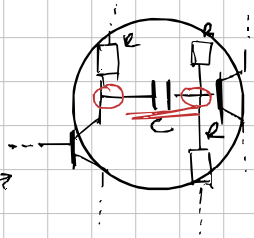
↳ R_{ul} >> → Bip T SZC

ali A_V < 1

↳ ... ugl nijedan nije savršen

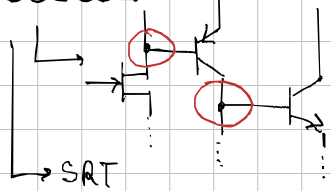
→ veća pojačanja pojačala uz različite ulazne i izlazne otpore mogu se postići ako se u složenom pojačalu kombinira više osnovnih spojeva pojačala => KASKADA POJAČALA

↳ serijski spojeni



Podjela:

- izmjenična pojačala → RC vezana (osigurava SRT stabilizaciju)
- istosmjerna pojačala => izravno vezana FET → Bip pnp → Bip npn
- Darlingtov spoj



↳ SRT
međusobno ovisne

(5.1) Kaskadno pojačalo

$R_{ul} >>$
$A_v >>$
$R_{iz} <<$

+ do 3 stupnja:

	SZE	SZB	SZC
R_{ul}	10–100	10–100	<1, ≈ 1
A_V	10–100	≈ 1, <1	10–100
A_I	$\sim k\Omega$	$\sim 10\Omega$	$\sim 10k\Omega$
R_{iz}	$\sim k\Omega$	$\sim 10\Omega$	$\sim 10\Omega$

→ spajamo međusobno da dobijemo tražena svojstva

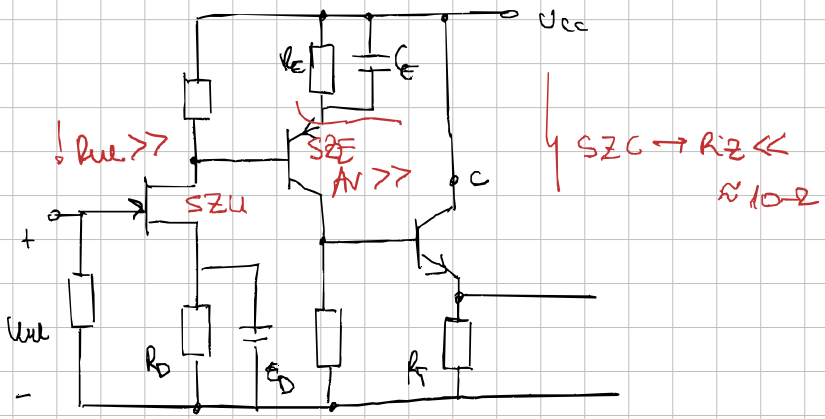
$$A_V = \prod_{k=1}^{N} A_{Vk}, \quad A_I = \prod_{k=1}^{N} A_{Ik}$$

dva načina → RC
→ izravno

SZE · SZC
↓ ↓
$R_{ul} >>$ $A_V >>$
$A_V >>$ $R_{iz} <<$

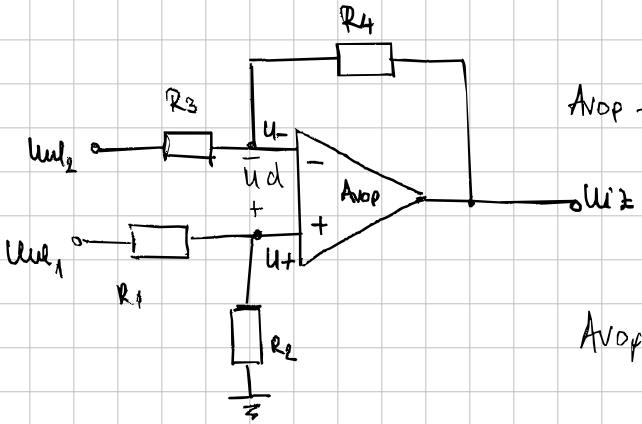
1. tješimo R_{ul} i R_{iz}
↓ ↓
FET SZC $\approx 10\Omega$

2. trebamo veliki A_V → SZE ≈ 100–200



3. GRUPA

1. Diferencijalno pojačalo s idealnim OP → pojačanje (Av) shema



$A_{vop} \to \infty$, $R_{ul} \to \infty$

$i_- = i_+ = \emptyset$ (neće struja u „teško")

$A_{vop}(u_+ - u_-) = u_{iz}$

$\Longrightarrow u_+ = u_-$

$\parallel$

u

1) $u\left(\frac{1}{R_3} + \frac{1}{R_4}\right) - \frac{1}{R_3} u_{ul2} - \frac{1}{R_4} u_{iz} = \emptyset$

2) $u\left(\frac{1}{R_1} + \frac{1}{R_2}\right) - \frac{1}{R_1} \cdot u_{ul1} = \emptyset$ $\qquad u = \frac{1}{R_1} \cdot u_{ul1} \cdot \left(\frac{1}{R_1} + \frac{1}{R_2}\right)^{-1} = u_{ul1} \cdot \frac{R_2}{R_1 + R_2}$

$$u_{ul1} \cdot \frac{R_2}{R_1 + R_2} \cdot \left(\frac{R_3 + R_4}{R_3 \cdot R_4}\right) - \frac{1}{R_3} u_{ul2} = \frac{1}{R_4} \cdot u_{iz} \quad \Big/ \cdot R_4$$

$$\boxed{u_{iz} = \underbrace{u_{ul1} \cdot \frac{R_2}{R_1 + R_2} \cdot \left(1 + \frac{R_4}{R_3}\right)}_{\langle \text{neinv. ulaz} \rangle \cdot \langle \text{pojačanje neinv. pojačala} \rangle} - \underbrace{\frac{R_4}{R_3} \cdot u_{ul2}}_{\langle \text{pojačanje inv. pojačala} \rangle \cdot \langle \text{napon na ulazu inv.} \rangle}}$$

↳ ovo pojačalo ima konačno R_{ul} → nije baš dobro

$u_d = u_+ - u_- \longrightarrow \frac{u_{iz}}{A_{vop}} = u_d$; $A_v = \frac{u_{iz}}{u_d} = A_{vop}$?

4. Pojačala snage

→ do sada su opisivana signalna pojačala

↳ uloga: pojačavanje signala; NAPONA ili STRUJA

▸ uz veće naponsko ili strujno pojačanje, bitna linearnost

! razine snaga izl. signala redovito male

▸ IZLAZ složenih pojačala:

- izlazni stupanj treba osigurati izlazni signal značajnog iznosa snage (predaje ga trošilu, npr. audiozvučniku)

IZLAZNA pojačala koja rade s većim snagama = POJAČALA SNAGE

▸ ne rade u linearnom režimu jer rade s većim signalima

↳ LINEARNOST je jedno od ključnih svojstava pojačala snage

- projektiranjem se nastoje smanjiti izobličenja

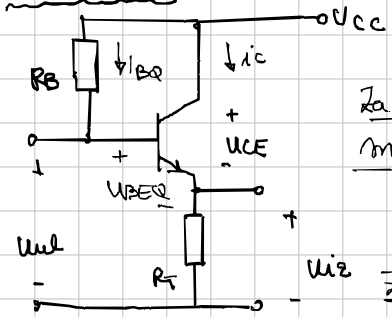
KLASA A

$I_{CQ} \geq i_{cm}$

tranzistor radi u NAP-u

▷ valni oblik izlaznog signala odgovara valnom obliku ulaz.

Klasa A



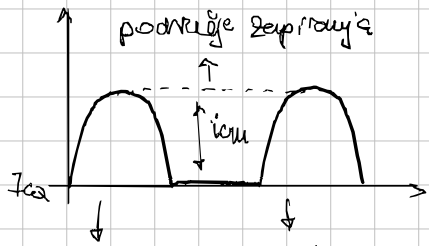
Za dobar prijenos snage: $R_{iz} \approx R_T \approx 4\,\Omega$ → poželjno da je R_{iz} mali

↳ BipT, SZD

Za trošila manjeg otpora: SZE

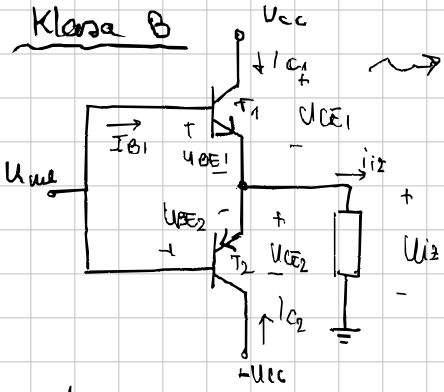
KLASA B

→ ne postavlja u SPT



tranz. radi u NAP-u u samo jednoj poz poluv.

Klasa B



→ Protutaktno pojačalo snage klase B

- problem neprovodnih odnosa snaga i male djelotvornosti A rješava se pomoću B

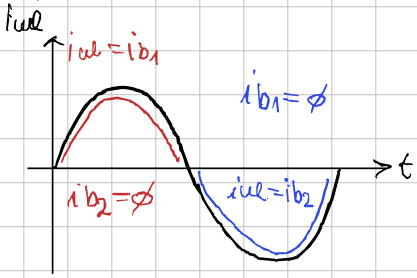
Djelotvornost ↑↑ uklanjanjem istosmjerne snage

- želimo izbjeći DC struju ⇒ T ne postavljamo u SRT

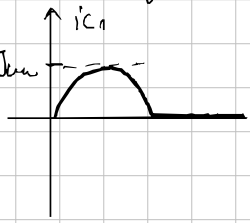
↳ uvijek vodi samo 1 od tranzistora

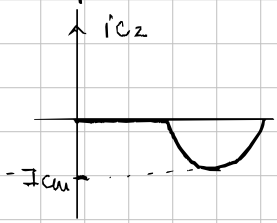
$i_{ul} = i_{B1} + i_{B2}$

npn radi na poz ulaz (propušta)
pnp radi na neg ulaz (propušta)



⇒

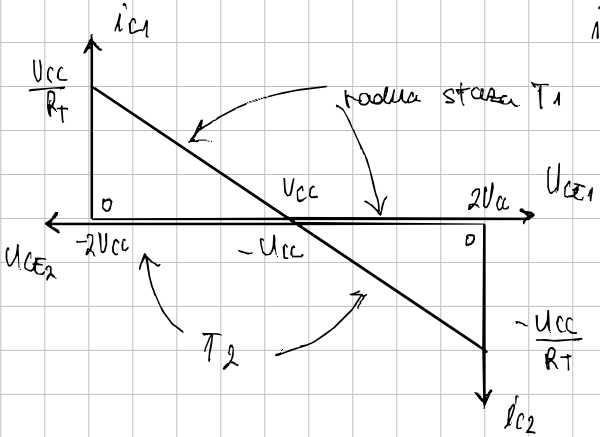




$$U_{iz} = i_{iz} \cdot R_T = (i_{C1} + i_{C2}) R_T$$

↳ posljedica komplementarnog para izvora napajanja i tranzistora
(U_{CC} i $-U_{CC}$) (npn i pnp)

Radne staze:



$i_{ul} > 0$ → vodi T_1, T_2 zapire
(NAP)

(NAP)
$i_{ul} < 0$ → vodi T_2, T_1 zapire